Hands-on Exercise No. 2 DigiSkills 3.0 Batch-04 Full Stack Development with MERN	Total Marks: 10 Due Date: 17/09/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
- It is submitted after due date.
- It is not in the required format (.doc or .docx)
- It does not open, or file is corrupt.
- It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this exercise, you shall be able to:

- Apply JavaScript variables, arrays, objects, functions, loops, and conditional statements to solve a simple product/order problem.
- Build a ReactJS frontend using components, props, state, event handling, list rendering, and CSS/TailwindCSS styling.
- Prepare a proper solution file by adding screenshots of code, terminal/browser output, and final ReactJS frontend output.

Problem Statement

In this hands-on exercise, you are required to apply your understanding of JavaScript problem solving and ReactJS frontend development. You will first solve a small product/order calculation problem using JavaScript. After that, you will build a simple ReactJS Product Catalog frontend using components, props, state, and styling.

The objective is to strengthen your logic-building skills and understand how JavaScript concepts are used inside ReactJS frontend applications.

You can use any code editor such as VS Code. For JavaScript, you may use the browser console or Node.js. For ReactJS, you may create the project using Vite. After completing each part, take screenshots of your code and output, and paste them into a single MS Word (.doc/.docx) file as your final solution submission.

Tasks:

Question 1: JavaScript Problem Solving - Product Order Calculator (Marks 5)

Create a JavaScript program using the given product data:

```
const products = [
  { id: 1, name: "Laptop", category: "Electronics", price: 85000, stock: 5 },
  { id: 2, name: "Mouse", category: "Accessories", price: 1500, stock: 0 },
  { id: 3, name: "Keyboard", category: "Accessories", price: 3500, stock: 10 },
  { id: 4, name: "Monitor", category: "Electronics", price: 25000, stock: 3 },
  { id: 5, name: "USB Cable", category: "Accessories", price: 700, stock: 15 },
];
```

- Part 1: Create a function that displays only available products whose stock is greater than 0. (Marks 2)
- Part 2: Create a function that calculates order subtotal, discount, and final total from cart items. Apply 10% discount if subtotal is greater than Rs. 50,000. (Marks 2)
- Part 3: Display a clear invoice/order summary in the console. (Marks 1)

Question 2: ReactJS Frontend Design - Product Catalog Page (Marks 5)

- Part 1: Create a ReactJS project using Vite and create a ProductCard component. The component should display product name, category, price, stock status, and an Add to Cart button. (Marks 2)
- Part 2: Display at least 4 product cards on the main page using an array and map() method. Apply CSS or TailwindCSS to create a clean and responsive grid layout. (Marks 2)
- Part 3: Use React state to show total cart items. When the user clicks Add to Cart, the cart count should increase. The button should be disabled for out-of-stock products. (Marks 1)

NOTE: Save the code file at your end and only provide screenshots of each step in MS Word File.

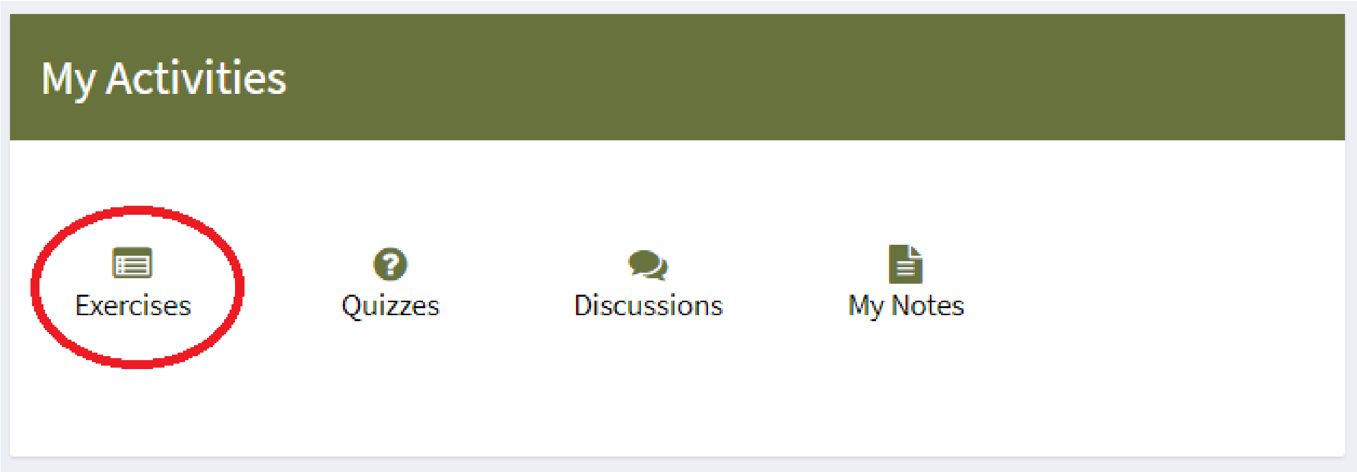
BEST OF LUCK :)

How to submit solution file on LMS?

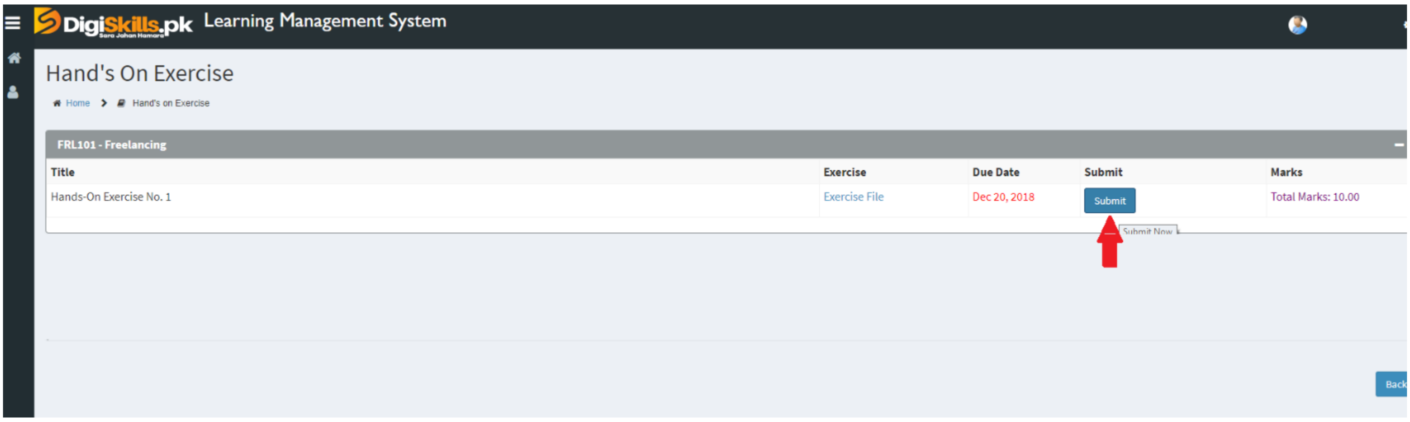
Please perform the following steps for submitting your solution using LMS:

1) Login to the LMS

2) Click on the **Exercises** button within the **My Activities** section



3) Click on the submit button to upload your Solution.



4) Keep in mind to upload your Solution in .doc or .docx format